Now Energy Can Be Harvested From Shadows

Scientists have come up with a new device called shadow-effect energy generator (SEG), which makes use of the contrast in illumination to harvest energy under weak ambient light. This article covers the architecture and application of SEG in energy generation

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Shadows are everywhere, and not much engineering use has been found for these. In conventional photovoltaic or optoelectronic applications where a steady source of light is used to power devices, the presence of shadows is undesirable since it degrades the performance of devices. But scientists have come up with a new device called shadow-effect energy generator (SEG), which makes use of the contrast in illumination that arises on the device from shadow castings to generate a direct current at 1.2V.

This device is capable of harvesting energy from illumination contrast arising under weak ambient light. Without any optimisation, SEG has a power density of 0.14 microwatt/square cm under indoor conditions of 0.001 suns where shadows are persistent. It performs 200 per cent better than commercial silicon solar cells under the effect of shadows.

The cost of components used to make SEG is around thirty to fifty dollars per square metre, compared with the hundreds of dollars used to make commercial solar panels. An SEG device with multiple generators in operation is shown in Fig. 1.

Architecture

Inside the shadow-effect energy generator, there is a set of silicon wafers that are arranged in a flexible and transparent plastic film. Each silicon wafer is deposited with an ultra-thin gold coating that is only 15nm thick (for reference, human DNA is about 2.5nm thick). When light meets silicon, photons or light particles knock electrons free from their atoms, which creates an energy flow.

Fig. 1: SEG in operation (Credit: Royal Society of Chemistry)

This is exactly the same process that a solar cell follows, but the gold layer is the true standout. With the gold coating, the device brings a stronger electric current when it is partially obscured in a shadow. As the agitated electrons jump from the silicon to the gold layer, the metal's voltage increases when there is a difference between the light and dark portions. The optimum surface area for electricity generation is when half of the SEG is illuminated, and the other half is in the shadow as this gives enough area for charge generation and collection, respectively.

Plasmonics for photovoltaics

Plasmonics is the name given in 2000 to the discipline for exploiting resonant interaction obtained under certain conditions between electromagnetic radiation (light in particular) and free electrons at the interface between metal and dielectric material like air or glass. This interaction generates electron density waves called plasmons or surface plasmons. One of the most promising ways to enhance the localised light absorption and improve the efficiency of thin solar wafers is to use plasmonic structure. Plasmons are embedded metal nanostructures that can localise incident light on a sub-micrometric scale, enabling light concentration and trapping.

Conventionally, photovoltaic absorbers must be optically thick to allow near-complete light absorption and photocarrier current collection. Fig. 2 shows the standard AM1.5 solar spectrum together with a graph that illustrates what fraction of the solar spectrum is absorbed on a single pass through a 2μm thick crystalline silicon film.